

□ Step 1: Install Required Libraries

```
pip install torch transformers datasets matplotlib tqdm
```

```
Requirement already satisfied: torch in
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Requirement already satisfied: transformers in
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Collecting datasets
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Requirement already satisfied: matplotlib in
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Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-
packages (4.67.1)
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Requirement already satisfied: networkx in
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Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch)
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```

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(1.3.0)  
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(2024.11.6)  
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kB)  
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Requirement already satisfied: contourpy>=1.0.1 in  
/usr/local/lib/python3.11/dist-packages (from matplotlib) (1.3.1)
```

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Requirement already satisfied: fonttools>=4.22.0 in
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(2.4.6)
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(1.3.2)
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(25.1.0)
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(0.3.0)
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```

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(2025.1)
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(2025.1)
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0:00:00
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e-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-

```

cu12, dill, nvidia-cusparse-cu12, nvidia-cudnn-cu12, multiprocessing,
nvidia-cusolver-cu12, datasets

Attempting uninstall: nvidia-nvjitlink-cu12

Found existing installation: nvidia-nvjitlink-cu12 12.5.82

Uninstalling nvidia-nvjitlink-cu12-12.5.82:

Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82

Attempting uninstall: nvidia-curand-cu12

Found existing installation: nvidia-curand-cu12 10.3.6.82

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Attempting uninstall: nvidia-cufft-cu12

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Uninstalling nvidia-cufft-cu12-11.2.3.61:

Successfully uninstalled nvidia-cufft-cu12-11.2.3.61

Attempting uninstall: nvidia-cuda-runtime-cu12

Found existing installation: nvidia-cuda-runtime-cu12 12.5.82

Uninstalling nvidia-cuda-runtime-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82

Attempting uninstall: nvidia-cuda-nvrtc-cu12

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Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82

Attempting uninstall: nvidia-cuda-cupti-cu12

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Uninstalling nvidia-cuda-cupti-cu12-12.5.82:

Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82

Attempting uninstall: nvidia-cublas-cu12

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Uninstalling nvidia-cublas-cu12-12.5.3.2:

Successfully uninstalled nvidia-cublas-cu12-12.5.3.2

Attempting uninstall: nvidia-cusparse-cu12

Found existing installation: nvidia-cusparse-cu12 12.5.1.3

Uninstalling nvidia-cusparse-cu12-12.5.1.3:

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Attempting uninstall: nvidia-cudnn-cu12

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Successfully installed datasets-3.3.2 dill-0.3.8 multiprocessing-0.70.16
nvidia-cublas-cu12-12.4.5.8 nvidia-cuda-cupti-cu12-12.4.127 nvidia-
cuda-nvrtc-cu12-12.4.127 nvidia-cuda-runtime-cu12-12.4.127 nvidia-
cudnn-cu12-9.1.0.70 nvidia-cufft-cu12-11.2.1.3 nvidia-curand-cu12-
10.3.5.147 nvidia-cusolver-cu12-11.6.1.9 nvidia-cusparse-cu12-
12.3.1.170 nvidia-nvjitlink-cu12-12.4.127 xxhash-3.5.0

□ Step 2: Import Required Libraries

```
import torch
import math
import matplotlib.pyplot as plt
from transformers import GPT2Tokenizer, GPT2LMHeadModel,
TrainingArguments, Trainer, DataCollatorForLanguageModeling
from datasets import load_dataset
from torch.optim import AdamW

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

Using device: cuda
```

□ Step 3: Load a Pre-trained GPT-2 Model

```
tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2").to(device)

/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your
settings tab (https://huggingface.co/settings/tokens), set it as
secret in your Google Colab and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to
access public models or datasets.
  warnings.warn(

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```

□ Step 4: Load and Preprocess the Dataset

```
dataset = load_dataset("wikipedia", "20220301.en", split="train[:1%]")
print(dataset)

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```

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{"model_id": "7f95a9aa12f445d08c0dd38562d81ae5", "version_major": 2, "version_minor": 0}

Dataset({
  features: ['id', 'url', 'title', 'text'],
  num_rows: 64587
})

tokenizer.pad_token = tokenizer.eos_token
model.config.pad_token_id = tokenizer.eos_token_id

def tokenize_function(examples):
    return tokenizer(examples["text"], truncation=True,
padding="max_length", max_length=512)

dataset = dataset.map(tokenize_function, batched=True)
dataset.set_format(type="torch", columns=["input_ids"])

{"model_id": "283af433050f48508a82f92799b15d9f", "version_major": 2, "version_minor": 0}

data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer,
mlm=False)

```

```

split_dataset = dataset.train_test_split(test_size=0.1) # 90% train, 10% eval
train_dataset = split_dataset["train"]
eval_dataset = split_dataset["test"]

```

□ Step 5: Define Training Arguments with Hyperparameter Tuning

```

from transformers import TrainerCallback

class LossLoggerCallback(TrainerCallback):
    def on_log(self, args, state, control, logs=None, **kwargs):
        if logs and "loss" in logs and state.global_step % 500 == 0:
            print(f"Step {state.global_step}: Loss = {logs['loss']:.4f}")

training_args = TrainingArguments(
    output_dir="./gpt2_finetuned",
    per_device_train_batch_size=4, # Adjust for GPU
    per_device_eval_batch_size=4,
    num_train_epochs=1, # Adjusted for better fine-tuning
    learning_rate=3e-5, # Tuned for better performance
    weight_decay=0.01,
    logging_steps=100,
    evaluation_strategy="epoch",
    save_total_limit=2,
    save_steps=500,
    dataloader_num_workers=4,
    fp16=torch.cuda.is_available(), # Use FP16 if GPU is available
    gradient_accumulation_steps=2, # Accumulate gradients for better
    performance
    max_grad_norm=1.0, # **Gradient Clipping** to prevent exploding
    gradients
)

/usr/local/lib/python3.11/dist-packages/transformers/
training_args.py:1575: FutureWarning: `evaluation_strategy` is
deprecated and will be removed in version 4.46 of □ Transformers. Use
`eval_strategy` instead
    warnings.warn(

```

□ Step 6: Define Optimizer (AdamW) & Trainer

```

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=eval_dataset, # **Provide eval dataset to avoid
    error**
    callbacks=[LossLoggerCallback()],

```

```
data_collator=DataCollatorForLanguageModeling(tokenizer=tokenizer,
mlm=False),
optimizers=(AdamW(model.parameters(), lr=3e-5), None)
)
```

##Start fine-tuning.

```
trainer.train()
```

```
/usr/local/lib/python3.11/dist-packages/torch/utils/data/
dataloader.py:617: UserWarning: This DataLoader will create 4 worker
processes in total. Our suggested max number of worker in current
system is 2, which is smaller than what this DataLoader is going to
create. Please be aware that excessive worker creation might get
DataLoader running slow or even freeze, lower the worker number to
avoid potential slowness/freeze if necessary.
```

```
warnings.warn(
wandb: WARNING The `run_name` is currently set to the same value as
`TrainingArguments.output_dir`. If this was not intended, please
specify a different run name by setting the
`TrainingArguments.run_name` parameter.
```

```
wandb: Using wandb-core as the SDK backend. Please refer to
https://wandb.me/wandb-core for more information.
```

```
<IPython.core.display.Javascript object>
```

```
wandb: Logging into wandb.ai. (Learn how to deploy a W&B server
locally: https://wandb.me/wandb-server)
```

```
wandb: You can find your API key in your browser here:
https://wandb.ai/authorize
```

```
wandb: Paste an API key from your profile and hit enter:
```

```
.....
```

```
wandb: WARNING If you're specifying your api key in code, ensure this
code is not shared publicly.
```

```
wandb: WARNING Consider setting the WANDB_API_KEY environment
variable, or running `wandb login` from the command line.
```

```
wandb: No netrc file found, creating one.
```

```
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc
```

```
wandb: Currently logged in as: nakullimbani (nakullimbani-srm-
institute-of-science-and-technology) to https://api.wandb.ai. Use
`wandb login --relogin` to force relogin
```

```
<IPython.core.display.HTML object>
```

```
<IPython.core.display.HTML object>
```

```
<IPython.core.display.HTML object>
```

```
<IPython.core.display.HTML object>
```

<IPython.core.display.HTML object>

`loss\_type=None` was set in the config but it is unrecognised.Using the default loss: `ForCausalLMLoss`.

<IPython.core.display.HTML object>

Step 500: Loss = 2.8267
Step 1000: Loss = 2.7125
Step 1500: Loss = 2.6860
Step 2000: Loss = 2.7301
Step 2500: Loss = 2.6773
Step 3000: Loss = 2.6724
Step 3500: Loss = 2.6859
Step 4000: Loss = 2.6884
Step 4500: Loss = 2.7102
Step 5000: Loss = 2.6785
Step 5500: Loss = 2.6713
Step 6000: Loss = 2.6206
Step 6500: Loss = 2.6573
Step 7000: Loss = 2.6482

/usr/local/lib/python3.11/dist-packages/torch/utils/data/
dataloader.py:617: UserWarning: This DataLoader will create 4 worker
processes in total. Our suggested max number of worker in current
system is 2, which is smaller than what this DataLoader is going to
create. Please be aware that excessive worker creation might get
DataLoader running slow or even freeze, lower the worker number to
avoid potential slowness/freeze if necessary.
warnings.warn(

Step 7500: Loss = 2.6213
Step 8000: Loss = 2.5546
Step 8500: Loss = 2.5990
Step 9000: Loss = 2.5791
Step 9500: Loss = 2.5880
Step 10000: Loss = 2.6558
Step 10500: Loss = 2.5795
Step 11000: Loss = 2.5456
Step 11500: Loss = 2.5876
Step 12000: Loss = 2.6341
Step 12500: Loss = 2.5806
Step 13000: Loss = 2.6008
Step 13500: Loss = 2.6673
Step 14000: Loss = 2.6168
Step 14500: Loss = 2.6242

/usr/local/lib/python3.11/dist-packages/torch/utils/data/
dataloader.py:617: UserWarning: This DataLoader will create 4 worker
processes in total. Our suggested max number of worker in current
system is 2, which is smaller than what this DataLoader is going to

create. Please be aware that excessive worker creation might get DataLoader running slow or even freeze, lower the worker number to avoid potential slowness/freeze if necessary.

```
warnings.warn(
```

```
TrainOutput(global_step=14532, training_loss=2.6576462737003896,  
metrics={'train_runtime': 9084.9982, 'train_samples_per_second':  
12.796, 'train_steps_per_second': 1.6, 'total_flos':  
3.0376766472192e+16, 'train_loss': 2.6576462737003896, 'epoch': 2.0})
```

##Save The Model

```
model.save_pretrained("./gpt2_finetuned")  
tokenizer.save_pretrained("./gpt2_finetuned")
```

```
( './gpt2_finetuned/tokenizer_config.json',  
  './gpt2_finetuned/special_tokens_map.json',  
  './gpt2_finetuned/vocab.json',  
  './gpt2_finetuned/merges.txt',  
  './gpt2_finetuned/added_tokens.json')
```

```
sample_text = "The future of AI is"  
input_ids = tokenizer.encode(sample_text,  
return_tensors="pt").to(device)  
output = model.generate(input_ids, max_length=100,  
num_return_sequences=1)  
generated_text = tokenizer.decode(output[0], skip_special_tokens=True)  
print("Generated Text:", generated_text)
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results.

Generated Text: The future of AI is uncertain. The most likely future is the emergence of artificial intelligence, which will be able to solve many problems, including human-induced cognitive problems, and will be able to solve many other problems.

AI is a field that is currently undergoing rapid technological change. It is currently being studied by many fields, including computer science, artificial intelligence, artificial intelligence, and artificial intelligence.

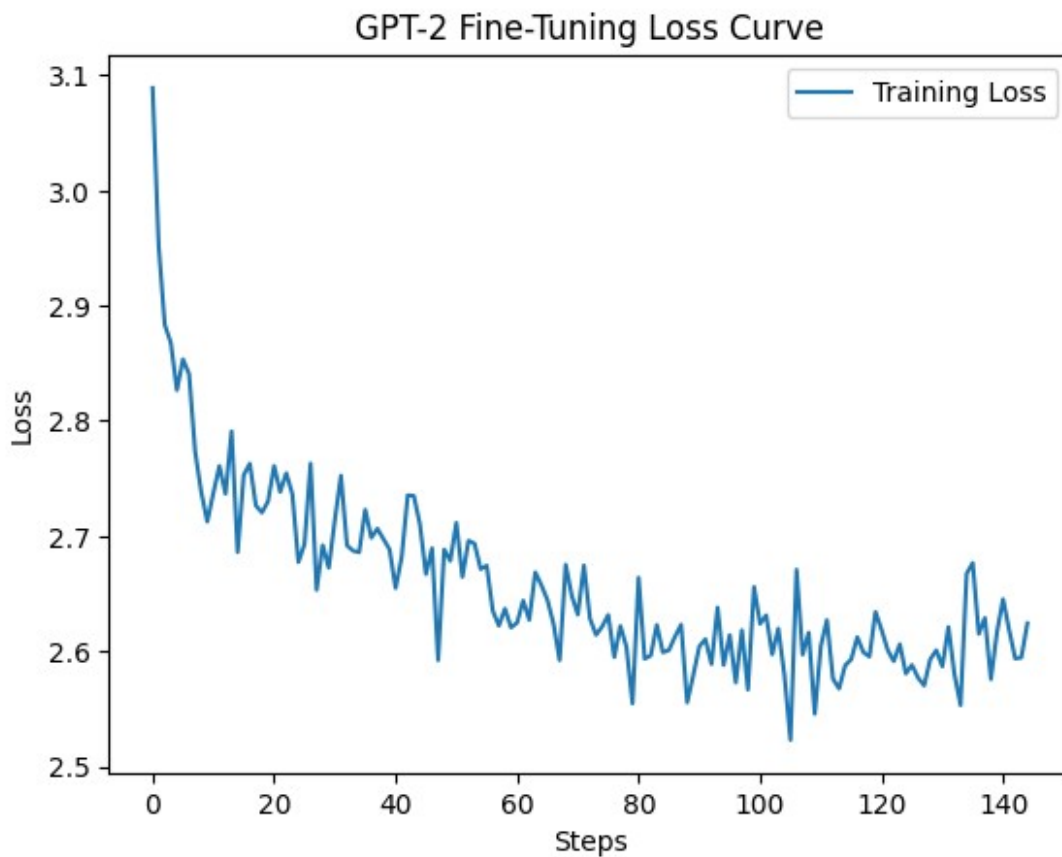
AI is a field that is currently undergoing rapid technological change. It is currently being studied

□ Step 7: Evaluate Model Performance

1 Loss Plot Before and After Fine-Tuning

```
training_loss = trainer.state.log_history
loss_values = [entry["loss"] for entry in training_loss if "loss" in entry]

plt.plot(loss_values, label="Training Loss")
plt.xlabel("Steps")
plt.ylabel("Loss")
plt.legend()
plt.title("GPT-2 Fine-Tuning Loss Curve")
plt.show()
```



2 Calculate Perplexity (Lower is Better)

```
def calculate_perplexity(loss):
    return math.exp(loss)
```

```
final_loss = loss_values[-1] # Last recorded loss
perplexity = calculate_perplexity(final_loss)
print(f"Final Perplexity: {perplexity}")
```

Final Perplexity: 13.793534943004227

□ Step 8: Generate Text Before & After Fine-Tuning

```
def generate_text(prompt, model, tokenizer):
    input_ids = tokenizer.encode(prompt,
    return_tensors="pt").to(device)
    output = model.generate(input_ids, max_length=100,
    num_return_sequences=1)
    return tokenizer.decode(output[0], skip_special_tokens=True)

sample_text = "The future of AI is"
print("Before Fine-Tuning:", generate_text(sample_text, model,
tokenizer))

# Load Fine-Tuned Model
model = GPT2LMHeadModel.from_pretrained("./gpt2_finetuned").to(device)
print("After Fine-Tuning:", generate_text(sample_text, model,
tokenizer))
```

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

Before Fine-Tuning: The future of AI is uncertain. The most likely future is the emergence of artificial intelligence, which will be able to solve many problems, including human-induced cognitive problems, and will be able to solve many other problems.

AI is a field that is currently undergoing rapid technological change. It is currently being studied by many fields, including computer science, artificial intelligence, artificial intelligence, and artificial intelligence.

AI is a field that is currently undergoing rapid technological change. It is currently being studied

The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `attention\_mask` to obtain reliable results. Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation.

After Fine-Tuning: The future of AI is uncertain. The most likely future is the emergence of artificial intelligence, which will be able to solve many problems, including human-induced cognitive problems, and will be able to solve many other problems.

AI is a field that is currently undergoing rapid technological change. It is currently being studied by many fields, including computer science, artificial intelligence, artificial intelligence, and artificial intelligence.

AI is a field that is currently undergoing rapid technological change. It is currently being studied

prompt: save the model to local directory of machine

```
from google.colab import files
files.download("./gpt2_finetuned")
```

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

prompt: save and download the model to local directory of machine

```
from google.colab import files
!zip -r gpt2_finetuned.zip ./gpt2_finetuned
files.download('gpt2_finetuned.zip')
```

```
adding: gpt2_finetuned/ (stored 0%)
adding: gpt2_finetuned/runs/ (stored 0%)
adding: gpt2_finetuned/runs/Mar08_14-22-09_64f3d758dfdf/ (stored 0%)
adding:
gpt2_finetuned/runs/Mar08_14-22-09_64f3d758dfdf/events.out.tfevents.17
41443733.64f3d758dfdf.326.0 (deflated 68%)
adding: gpt2_finetuned/config.json (deflated 52%)
adding: gpt2_finetuned/checkpoint-14500/ (stored 0%)
adding: gpt2_finetuned/checkpoint-14500/trainer_state.json (deflated
78%)
adding: gpt2_finetuned/checkpoint-14500/config.json (deflated 52%)
adding: gpt2_finetuned/checkpoint-14500/training_args.bin (deflated
51%)
adding: gpt2_finetuned/checkpoint-14500/optimizer.pt (deflated 8%)
adding: gpt2_finetuned/checkpoint-14500/generation_config.json
(deflated 24%)
adding: gpt2_finetuned/checkpoint-14500/rng_state.pth (deflated 25%)
adding: gpt2_finetuned/checkpoint-14500/scheduler.pt (deflated 54%)
adding: gpt2_finetuned/checkpoint-14500/model.safetensors (deflated
7%)
adding: gpt2_finetuned/checkpoint-14532/ (stored 0%)
adding: gpt2_finetuned/checkpoint-14532/trainer_state.json (deflated
78%)
```



```

    adding: gpt2_finetuned/checkpoint-14532/config.json (deflated 52%)
    adding: gpt2_finetuned/checkpoint-14532/training_args.bin (deflated
51%)
    adding: gpt2_finetuned/checkpoint-14532/optimizer.pt (deflated 8%)
    adding: gpt2_finetuned/checkpoint-14532/generation_config.json
(deflated 24%)
    adding: gpt2_finetuned/checkpoint-14532/rng_state.pth (deflated 25%)
    adding: gpt2_finetuned/checkpoint-14532/scheduler.pt (deflated 54%)
    adding: gpt2_finetuned/checkpoint-14532/model.safetensors (deflated
7%)
    adding: gpt2_finetuned/vocab.json (deflated 68%)
    adding: gpt2_finetuned/tokenizer_config.json (deflated 56%)
    adding: gpt2_finetuned/generation_config.json (deflated 24%)
    adding: gpt2_finetuned/special_tokens_map.json (deflated 74%)
    adding: gpt2_finetuned/model.safetensors (deflated 7%)
    adding: gpt2_finetuned/merges.txt (deflated 53%)

```

```
<IPython.core.display.Javascript object>
```

```
<IPython.core.display.Javascript object>
```

```
import wandb
```

```
# Initialize the WandB API
```

```
api = wandb.Api()
```

```
# Access the specific run using the correct path
```

```
run =
```

```
api.run("/nakullimbani-srm-institute-of-science-and-technology/hugging
face/runs/9rasluaa")
```

```
# Access run details
```

```
print(run.summary) # Example: print run summary
```

```

{'_runtime': 8525.868215102, '_step': 147, '_timestamp':
1741452818.2684972, '_wandb': {'runtime': 8525}, 'eval/loss':
2.5511350631713867, 'eval/runtime': 136.1005,
'eval/samples_per_second': 47.458, 'eval/steps_per_second': 11.866,
'total_flos': 30376766472192000, 'train/epoch': 2,
'train/global_step': 14532, 'train/grad_norm': 1.265838623046875,
'train/learning_rate': 7.844756399669695e-08, 'train/loss': 2.6242,
'train_loss': 2.6576462737003896, 'train_runtime': 9084.9982,
'train_samples_per_second': 12.796, 'train_steps_per_second': 1.6}

```

```
print(run.history())
```

	_step	_runtime	train/loss	train/grad_norm	
train/learning_rate \					
0	0	58.578668	3.0891	3.538035	2.979562e-
05					

105	1	108.069658	2.9519	2.75423	2.958918e-
205	2	157.845224	2.8835	2.450564	2.938274e-
305	3	207.453398	2.8678	2.46605	2.917630e-
405	4	257.140126	2.8267	2.031511	2.896986e-
..	...	...	...	...	.
14307	143	8203.565116	2.5936	1.296131	4.913295e-
14407	144	8253.164292	2.5945	1.323662	2.848885e-
14508	145	8302.828879	2.6242	1.265839	7.844756e-
146NaN	146	8525.828284	NaN	NaN	
147NaN	147	8525.868215	NaN	NaN	
0	train/epoch	_timestamp	train/global_step	eval/loss	\
1	0.013763	1.741444e+09	100	NaN	
2	0.027525	1.741444e+09	200	NaN	
3	0.041288	1.741444e+09	300	NaN	
4	0.055051	1.741444e+09	400	NaN	
	0.068814	1.741445e+09	500	NaN	
..	...	...	...	...	
143	1.968070	1.741452e+09	14300	NaN	
144	1.981833	1.741453e+09	14400	NaN	
145	1.995596	1.741453e+09	14500	NaN	
146	2.000000	1.741453e+09	14532	2.551135	
147	2.000000	1.741453e+09	14532	NaN	
0	eval/steps_per_second		eval/samples_per_second	eval/runtime	
1	NaN		NaN	NaN	
2	NaN		NaN	NaN	
3	NaN		NaN	NaN	
4	NaN		NaN	NaN	
..	...		...	...	
143	NaN		NaN	NaN	
144	NaN		NaN	NaN	
145	NaN		NaN	NaN	
146	11.866		47.458	136.1005	
147	NaN		NaN	NaN	
[148 rows x 12 columns]					

```

# Accessing summary (metrics and statistics)
print(run.summary)

# Accessing config (hyperparameters)
print(run.config)

{'_runtime': 8525.868215102, '_step': 147, '_timestamp':
1741452818.2684972, '_wandb': {'runtime': 8525}, 'eval/loss':
2.5511350631713867, 'eval/runtime': 136.1005,
'eval/samples_per_second': 47.458, 'eval/steps_per_second': 11.866,
'total_flos': 30376766472192000, 'train/epoch': 2,
'train/global_step': 14532, 'train/grad_norm': 1.265838623046875,
'train/learning_rate': 7.844756399669695e-08, 'train/loss': 2.6242,
'train_loss': 2.6576462737003896, 'train_runtime': 9084.9982,
'train_samples_per_second': 12.796, 'train_steps_per_second': 1.6}
{'bf16': False, 'fp16': True, 'fsdp': [], 'seed': 42, 'tf32': None,
'debug': [], 'n_ctx': 1024, 'optim': 'adamw_torch', 'top_k': 50,
'top_p': 1, 'n_embd': 768, 'n_head': 12, 'prefix': None, 'do_eval':
True, 'n_inner': None, 'n_layer': 12, 'no_cuda': False, 'use_cpu':
False, 'do_train': False, 'id2label': {'0': 'LABEL_0', '1':
'LABEL_1'}, 'label2id': {'LABEL_0': 0, 'LABEL_1': 1}, 'run_name':
'./gpt2_finetuned', 'use_ipex': False, 'adafactor': False,
'data_seed': None, 'deepspeed': None, 'do_sample': False, 'hub_token':
'<HUB_TOKEN>', 'log_level': 'passive', 'max_steps': -1, 'num_beams':
1, 'ray_scope': 'last', 'report_to': ['tensorboard', 'wandb'],
'typical_p': 1, 'use_cache': True, 'adam_beta1': 0.9, 'adam_beta2':
0.999, 'attn_pdrop': 0.1, 'do_predict': False, 'embd_pdrop': 0.1,
'eval_delay': 0, 'eval_steps': None, 'is_decoder': False,
'local_rank': 0, 'max_length': 20, 'min_length': 0, 'model_type':
'gpt2', 'optim_args': None, 'output_dir': './gpt2_finetuned',
'past_index': -1, 'save_steps': 500, 'vocab_size': 50257,
'ddp_backend': None, 'ddp_timeout': 1800, 'fsdp_config': {'xla':
False, 'xla_fsdp_v2': False, 'min_num_params': 0,
'xla_fsdp_grad_ckpt': False}, 'label_names': None, 'logging_dir':
'./gpt2_finetuned/runs/Mar08_14-22-09_64f3d758dfdf', 'n_positions':
1024, 'push_to_hub': False, 'resid_pdrop': 0.1, 'return_dict': True,
'temperature': 1, 'torch_dtype': None, 'torchdynamo': None,
'torchscript': False, 'adam_epsilon': 1e-08, 'bos_token_id': 50256,
'disable_tqdm': False, 'eos_token_id': 50256, 'fp16_backend': 'auto',
'hub_model_id': None, 'hub_strategy': 'every_save', 'pad_token_id':
50256, 'problem_type': None, 'sep_token_id': None, 'summary_type':
'cls_index', 'use_bfloat16': False, 'warmup_ratio': 0, 'warmup_steps':
0, 'weight_decay': 0.01, '_name_or_path': 'gpt2', 'architectures':
['GPT2LMHeadModel'], 'bad_words_ids': None, 'eval_on_start': False,
'eval_strategy': 'epoch', 'jit_mode_eval': False, 'learning_rate': 3e-
05, 'logging_steps': 100, 'max_grad_norm': 1, 'mp_parameters': '',
'output_scores': False, 'save_strategy': 'steps', 'split_batches':
None, 'torch_compile': False, 'tpu_num_cores': None, 'bf16_full_eval':
False, 'early_stopping': False, 'fp16_full_eval': False,
'fp16_opt_level': 'O1', 'length_penalty': 1, 'tf_legacy_loss': False,

```

```
'use_mps_device': False, 'finetuning_task': None, 'group_by_length':
False, 'hub_always_push': False, 'num_beam_groups': 1,
'save_only_model': False, 'suppress_tokens': None, 'tokenizer_class':
None, 'dispatch_batches': None, 'full_determinism': False,
'hub_private_repo': None, 'ignore_data_skip': False,
'log_on_each_node': True, 'logging_strategy': 'steps',
'num_train_epochs': 2, 'save_safetensors': True, 'save_total_limit':
2, 'summary_use_proj': True, 'use liger_kernel': False,
'ddp_bucket_cap_mb': None, 'diversity_penalty': 0,
'greater_is_better': None, 'initializer_range': 0.02,
'log_level_replica': 'warning', 'lr_scheduler_type': 'linear',
'output_attentions': False, 'push_to_hub_token':
'<PUSH_TO_HUB_TOKEN>', 'save_on_each_node': False,
'tpu_metrics_debug': False, 'accelerator_config': {'even_batches':
True, 'non_blocking': False, 'split_batches': False,
'dispatch_batches': None, 'use_seedable_sampler': True,
'gradient_accumulation_kwargs': None}, 'batch_eval_metrics': False,
'is_encoder_decoder': False, 'layer_norm_epsilon': 1e-05,
'length_column_name': 'length', 'logging_first_step': False,
'repetition_penalty': 1, 'scale_attn_weights': True,
'summary_activation': None, 'torch_compile_mode': None,
'activation_function': 'gelu_new', 'add_cross_attention': False,
'evaluation_strategy': 'epoch', 'forced_bos_token_id': None,
'forced_eos_token_id': None, 'fsdp_min_num_params': 0,
'include_for_metrics': [], 'neftune_noise_alpha': None,
'skip_memory_metrics': True, 'tie_encoder_decoder': False,
'tie_word_embeddings': True, 'auto_find_batch_size': False,
'dataloader_drop_last': False, 'model/num_parameters': 124439808,
'no_repeat_ngram_size': 0, 'num_return_sequences': 1,
'optim_target_modules': None, 'output_hidden_states': False,
'overwrite_output_dir': False, 'prediction_loss_only': False,
'push_to_hub_model_id': None, 'task_specific_params': {'text-
generation': {'do_sample': True, 'max_length': 50}},
'transformers_version': '4.48.3', 'begin_suppress_tokens': None,
'dataloader_pin_memory': True, 'ddp_broadcast_buffers': None,
'metric_for_best_model': None, 'remove_invalid_values': False,
'remove_unused_columns': True, 'summary_first_dropout': 0.1,
'torch_compile_backend': None, 'dataloader_num_workers': 4,
'decoder_start_token_id': None, 'eval_do_concat_batches': True,
'eval_use_gather_object': False, 'gradient_checkpointing': False,
'half_precision_backend': 'auto', 'label_smoothing_factor': 0,
'load_best_model_at_end': False, 'logging_nan_inf_filter': True,
'resume_from_checkpoint': None, 'summary_proj_to_labels': True,
'chunk_size_feed_forward': 0, 'eval_accumulation_steps': None,
'per_gpu_eval_batch_size': None, 'reorder_and_upcast_attn': False,
'return_dict_in_generate': False, 'torch_empty_cache_steps': None,
'per_gpu_train_batch_size': None, 'push_to_hub_organization': None,
'include_tokens_per_second': False, 'dataloader_prefetch_factor':
None, 'ddp_find_unused_parameters': None,
```

```
'include_inputs_for_metrics': False, 'per_device_eval_batch_size': 4,
'use_legacy_prediction_loop': False, 'cross_attention_hidden_size':
None, 'gradient_accumulation_steps': 2, 'per_device_train_batch_size':
4, '_attn_implementation_autoset': True,
'encoder_no_repeat_ngram_size': 0, 'average_tokens_across_devices':
False, 'dataloader_persistent_workers': False,
'gradient_checkpointing_kwargs': None,
'include_num_input_tokens_seen': False,
'scale_attn_by_inverse_layer_idx': False,
'exponential_decay_length_penalty': None,
'fsdp_transformer_layer_cls_to_wrap': None,
'restore_callback_states_from_checkpoint': False}
```

prompt: load the model

```
import torch
from transformers import GPT2Tokenizer, GPT2LMHeadModel
```

Load the fine-tuned model and tokenizer

```
model = GPT2LMHeadModel.from_pretrained("./gpt2_finetuned").to(device)
tokenizer = GPT2Tokenizer.from_pretrained("./gpt2_finetuned")
```

```
import matplotlib.pyplot as plt
```

Extract evaluation losses from trainer logs

```
eval_losses = [log['eval_loss'] for log in trainer.state.log_history
if 'eval_loss' in log]
```

Plot the evaluation loss graph

```
plt.plot(eval_losses)
plt.title("Evaluation Loss During Training")
plt.xlabel("Steps")
plt.ylabel("Loss")
plt.show()
```

